Introduction to Bayesian inference using brms and Stan

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Outline

1. Introduction: NHST vs. Bayesian inference

2. Bayesian modeling with brms

3. Example: t-test vs. equivalent Bayesian model

4. Conclusion

Bayesian Modeling with brms

Inference

- Going from a set of premises to a conclusion.

- Deduction: Conclusion is necessary.

- Induction: Conclusion is probable (where $0 > \text{probability} < 1$).

Statistical (Inductive) Inference

- Your model and data are your <u>premises</u>.
- Your conclusion is not <u>necessary</u> given your data (i.e. it relies on induction).
- Statistics let’s you quantify the strength of your conclusion.

NHST: Null-Hypothesis Significance Testing

- NHST is the dominant way to test hypotheses today.
- People who report and rely on ‘p values’ are using NHST.
- NHST is a bit of a convoluted framework.

NHST: Null-Hypothesis Significance Testing

- Formulate hypotheses:
  - H0 = Effect is zero
  - H1 = Effect is not zero

- Find the probability of a result as extreme as the one you observed (p-value) given H0.

- If $p < \alpha$ (usually $\alpha=0.05$) you reject H0 and accept H1.

NHST: Null-Hypothesis Significance Testing

- Note that NHST does <u>not</u>:
  - Tell you the probability that H1 is true.
  - Directly provide you probable values of parameters related to H1.
  - Allow you to say that H0 is true.

NHST: Coin Example

You pull a random coin out of your pocket. We want to know if the coin is ‘fair’, i.e., does flipping it result in 50% heads in the long run?

NHST: Hypotheses

We can view our question as a competition between two hypotheses regarding the probability of a heads result (p):

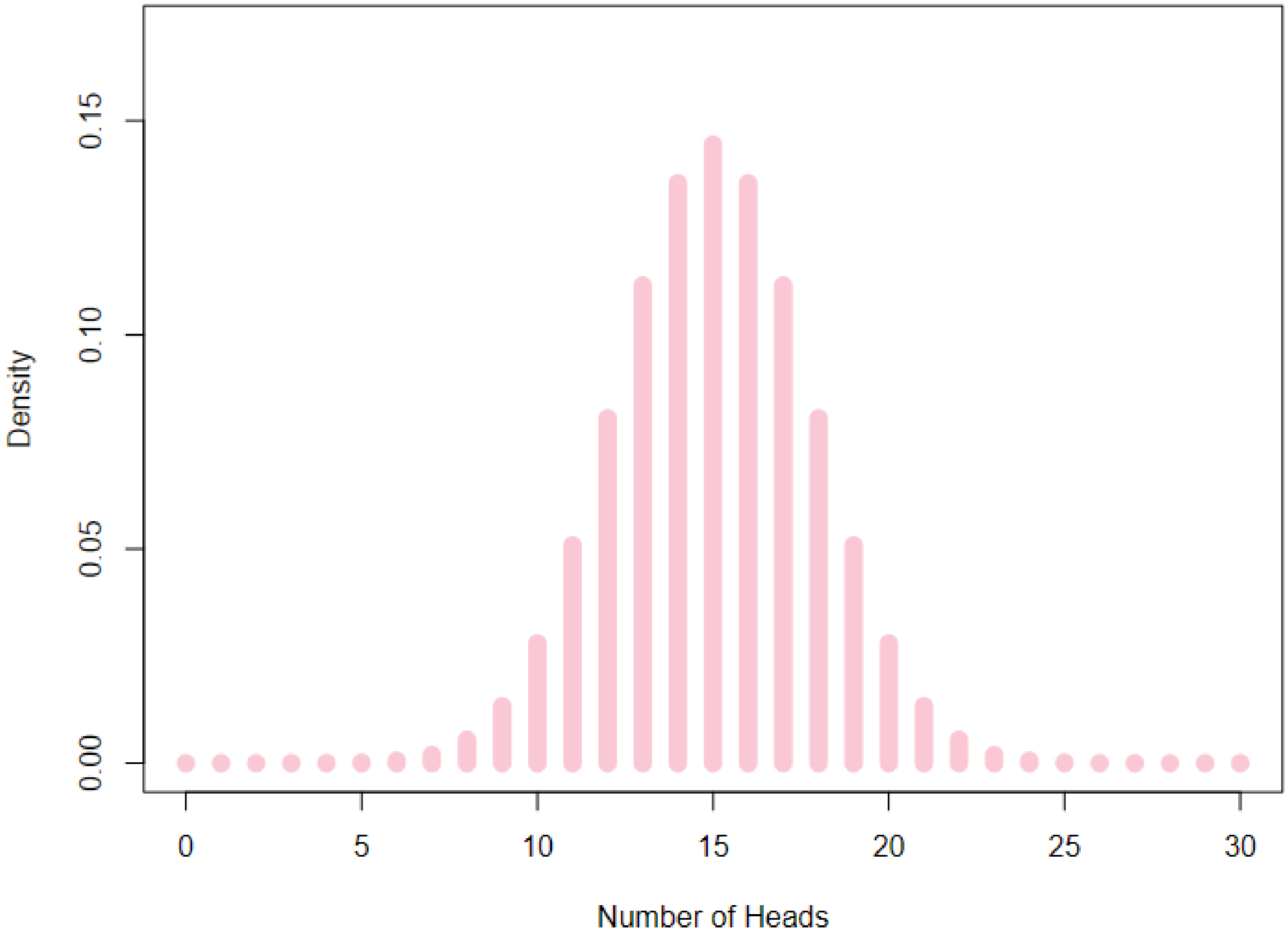
- H0 = The coin is fair, i.e., $p = 0.5$.
- H1 = The coin is not fair, i.e., $p \neq 0.5$.

NHST: The Data

- You flip it 30 times and see 21/30 heads, a rate of $p=0.7$ or 70% heads.

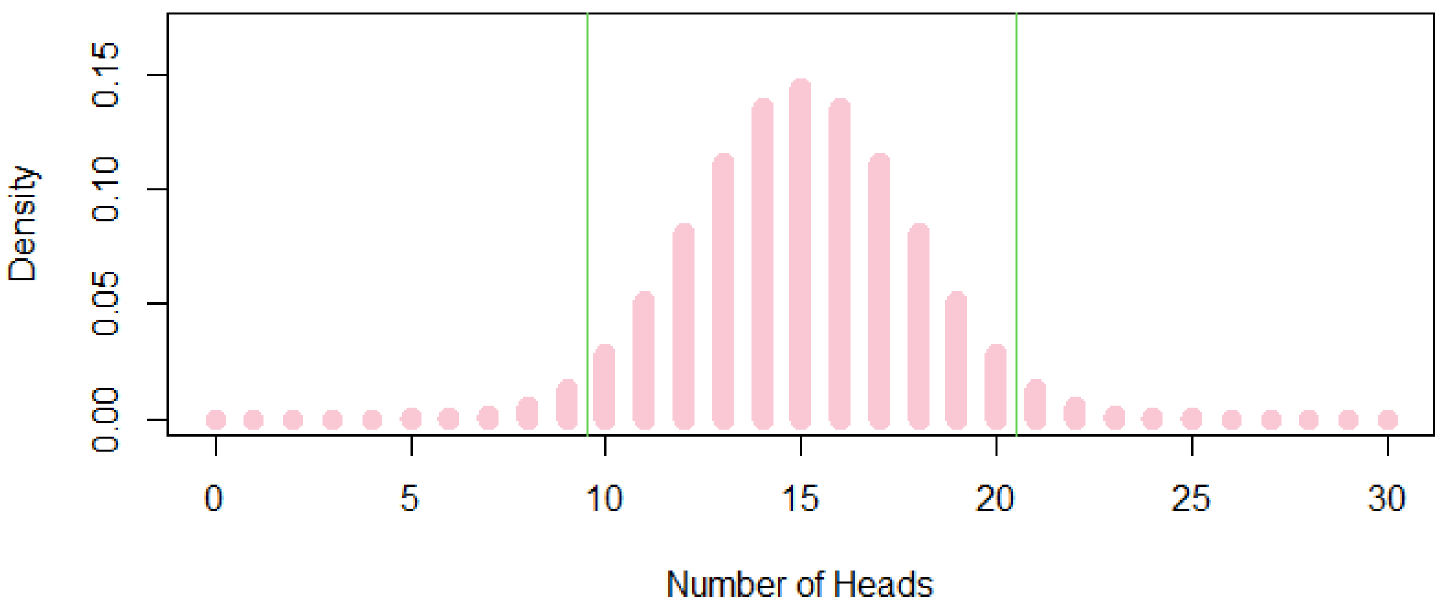
- This is your sample of data (D). The specific sequence does not matter so much as the overall rate of heads.

NHST: The Sampling Distribution



NHST: The Sampling Distribution

```
# probability of fewer than 10 successes
pbinom (9,30,.5)
## [1] 0.02138697

# probability of more than 21 successes
1 - pbinom (20,30,.5)
## [1] 0.02138697

# probability of fewer than 10 successes or more than 21 successes
pbinom (9,30,.5) + (1 - pbinom (20,30,.5))
## [1] 0.04277395
```



NHST: Rejecting the Null

- By convention, we reject the null hypothesis when $p < 0.05$.

- As a result, we conclude that the coin is not fair:

  - H0 = The coin is fair, i.e. rate = 0.5 ✕
  - H1 = The coin is not fair

NHST: Accepting the Alternative

- We accept the alternative hypothesis:
  - H0 = The coin is fair, i.e. rate = 0.5
  - H1 = The coin is not fair ✓
- We usually conclude that p=0.7. This is the value that makes our data most probable.

NHST: Confidence intervals

- Our confidence interval for p is 0.51 to 0.82.
- This is <u>not</u> an interval that contains p with 95% probability.
- Instead, under a large number of replications, this interval will contain p 95% of the time.

Bayesian Inference (BI)

- Formulate hypotheses (H):
  - H1 = Effect is zero
  - H2 = Effect is some other value

- Specify prior probabilities (or distributions) over the hypotheses or model parameters.

- Find the posterior probability of different hypotheses.

Bayes Theorem

$$P(A \& B) = P(B \& A)$$

$$P(A \mid B) \cdot P(B) = P(B \mid A) \cdot P(A)$$

$$P(\mu \mid y) \cdot P(y) = P(y \mid \mu) \cdot P(\mu)$$

$$\boxed{P(\mu \mid y) = \frac{P(y \mid \mu) \cdot P(\mu)}{P(y)}}$$

Bayes Theorem

- Two ways to think of Bayes theorem:

$$P(H|D) = \frac{P(D|H)P(H)}{P(D)} = P(H) \cdot \frac{P(D|H)}{P(D)}$$

- Scales the prior given how your data changes things:

$$P(\mathrm{H}|\mathrm{D}) = P(\mathrm{H}) \cdot \frac{P(\mathrm{D}|\mathrm{H})}{P(\mathrm{D})}$$

Prior Probabilities

- Probability of different values of your parameter prior to collecting data.

- A priori expectation comes from world knowledge, previous experiments, common sense, or some combination thereof.

$$P(\mu \mid y) = \frac{P(y \mid \mu) \cdot P(\mu)}{P(y)}$$

Likelihood

- The joint probability of your data given a value of μ.

$$P(\mu \mid y) = \frac{P(y \mid \mu) \cdot P(\mu)}{P(y)}$$

- Reflects the probability that the data, y, would be observed for values of μ.

NHST vs. BI

- ‘Traditional’ (frequentist) models focus mostly on the likelihoods of parameters to make inferences.

$$P(\mu \mid y) = \frac{P(y \mid \mu) \cdot P(\mu)}{P(y)}$$

- Bayesian models make inferences using the posterior probability of parameters.

Marginal/Total Probability

- Scale the numerator so that the posterior density has a total area under the curve equal to one.

$$P(\mu \mid y) = \frac{P(y \mid \mu) \cdot P(\mu)}{P(y)}$$

- You don't typically need to worry about it.

Posterior Probability

- The posterior probability of values of μ given your data y, and the structure of your model.

$$P(\mu \mid y) = \frac{P(y \mid \mu) \cdot P(\mu)}{P(y)}$$

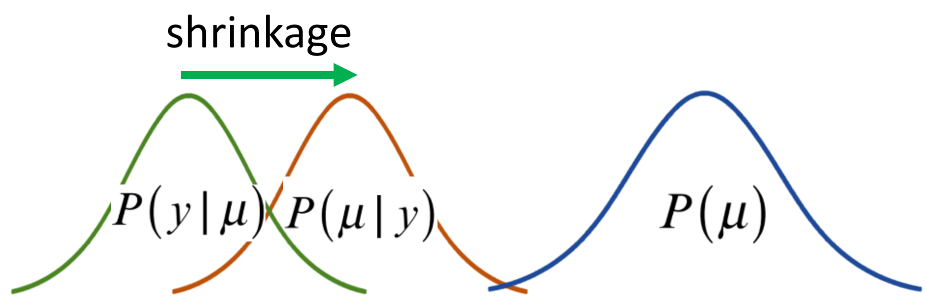
- A combination of the prior distribution and the likelihood, a combination of old and new information.

Posterior Distributions and Shrinkage

- Posterior probabilities can be pulled away from the most likely parameter values, towards the prior.

- This 'pull' is called *shrinkage*, values get *shrunk* towards the prior.

$$P(\mu \mid y) = \frac{P(y \mid \mu) \cdot P(\mu)}{P(y)}$$



Bayesian Models

- Bayesian models:

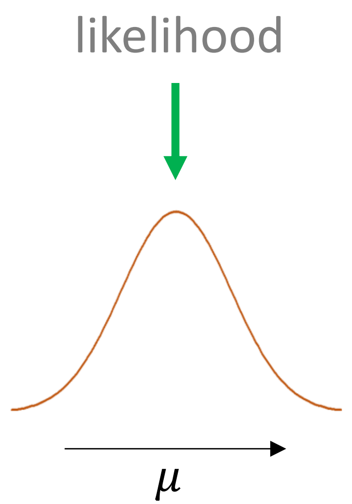
   1. Specify prior probabilities for all parameters: $P(\mu)$

   2. Base inference on posterior probabilities: $P(\mu|y)$

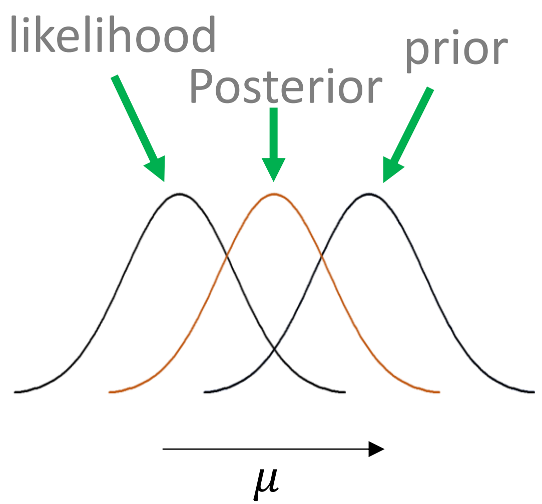
- Posterior probabilities are (basically) just likelihoods times priors: $P(y|\mu) \cdot P(\mu)$

- 'Traditional' models base inference on likelihoods only (i.e. ignore priors): $P(y|\mu)$

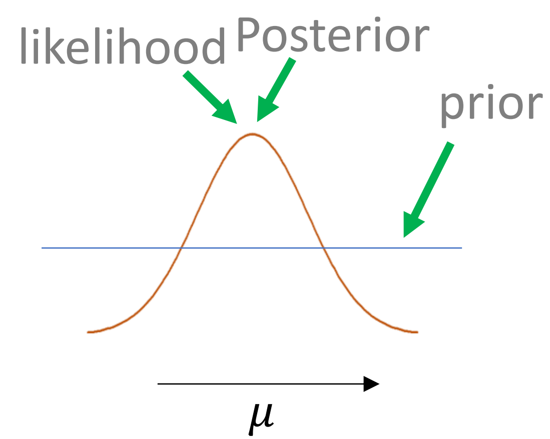
Posterior Distributions and Shrinkage



Maximum
likelihood
"Frequentist"



Posterior Mean
Bayesian



Posterior Mean
'Bayesian'

BI: Coin Flip Example

- We observe 21/30 heads on 30 coin-flips.
- Competing hypotheses:
  - H1 = The coin is fair, i.e. rate =0.5.
  - H2 = The coin is not fair, i.e. rate =0.7.

BI: Assigning Priors

- We can assume no prior knowledge:
  - H1_prior = 0.5
    H2_prior = 0.5

- Or we can assume knowledge of the world:
  - H1_prior = 0.999
    H2_prior = 0.001

BI: The Likelihood

- We can find the likelihood of the data for each hypothesis:

```
H1_likelihood = dbinom(21, 30, 0.5)
H1_likelihood
```

```
[1] 0.01332457
```

```
H2_likelihood = dbinom(21, 30, 0.7)
H2_likelihood
```

```
[1] 0.1572908
```

BI: Finding the Posterior

$$P\left(\mathrm{H}_j|\mathrm{Data}\right) = \frac{P\left(\mathrm{Data}|\mathrm{H}_j\right) \cdot P\left(\mathrm{H}_j\right)}{P(D)}$$

When we have only a fixed number of hypotheses, $P(D)$ can be calculated as seen below:

$$P\left(\mathrm{H}_j|\mathrm{Data}\right) = \frac{P\left(\mathrm{Data}|\mathrm{H}_j\right) \cdot P\left(\mathrm{H}_j\right)}{\sum_{i=1}^{2} P\left(\mathrm{Data}|\mathrm{H}_i\right) \cdot P(\mathrm{H}_i)}$$

BI: Finding the Posterior

$$P(\mathrm{H1}|\mathrm{Data}) = \frac{P(\mathrm{Data}|\mathrm{H1}) \cdot P(H1)}{P(D)}$$

$$= \frac{P(\mathrm{Data}|\mathrm{H1}) \cdot P(H1)}{P(\mathrm{Data}|\mathrm{H1}) \cdot P(H1) + P(\mathrm{Data}|\mathrm{H2}) \cdot P(H2)}$$

$$P(\mathrm{Data}|\mathrm{H1}) \cdot P(H1) = 0.0133 \cdot 0.999 = 0.0133$$
$$P(\mathrm{Data}|\mathrm{H2}) \cdot P(H2) = 0.1573 \cdot 0.001 = 0.0002$$
$$P(D) = 0.0133 + 0.002 = 0.0134$$

BI: Finding the Posterior

- Using this approach, $P(H1|D) = 0.993$, and $P(H2|D) = 0.007$.
- There is a probability of 0.993 that the coin is fair (H1), and a 0.007 probability that the coin is not fair, where $p=0.7$ (H2).
- We do not accept or reject one or the other, we get a direct measure of the credibility of hypotheses.

BI: Finding the Posterior

- What if we instead adopt the 'naïve' priors:

$$P(\text{Data}|\text{H1}) \cdot P(H1) = 0.0133 \cdot 0.5 = 0.0067$$
$$P(\text{Data}|\text{H2}) \cdot P(H2) = 0.1573 \cdot 0.5 = 0.0787$$
$$P(D) = 0.0067 + 0.1573 = 0.0853$$

- In this case, P(H1|D) = 0.078, and P(H2|D) = 0.922.

- So, there is a probability of 0.078 that the coin is fair, and 0.922 that it is not.

NHST .vs BI

- Based on the priors we use, we think:
  - There is a 0.078 probability that it is fair (if naïve a priori)
  - There is a 0.993 probability that it is fair (if not naïve a priori)
- What would you believe?

NHST .vs BI

- What if I offer you the following bet: We will flip the coin 100 more times:
    - If we get >60 heads I give you $100.
    - If we get <60 heads you give me $100.
- If you think p=0.7 you have a 98% chance of winning the bet.
- If you think p=0.5 you have a 2% chance of winning the bet.

NHST vs. BI: Conclusion

- In ‘real life’ we use prior information to make inferences.
- We don’t need to interpret data in isolation in science.
- Bayesian inference offers a principled way to convert old and new information.

Bayesian Modeling with brms

Sampling from the Posterior

- We need information about $P(\mathrm{H}|\mathrm{D})$.
- For complex models this can be difficult/impossible to estimate analytically.
- Instead, we can investigate $P(\mathrm{H}|\mathrm{D})$ with sampling (just like in your experiments).

Sampling Software

- Stan: a probabilistic programming language.
- brms: is an R package that writes Stan models for you and provides many useful helper functions.
- We can use Stan and brms to understand the posterior distribution of our model parameters.

Sampling from a Distribution

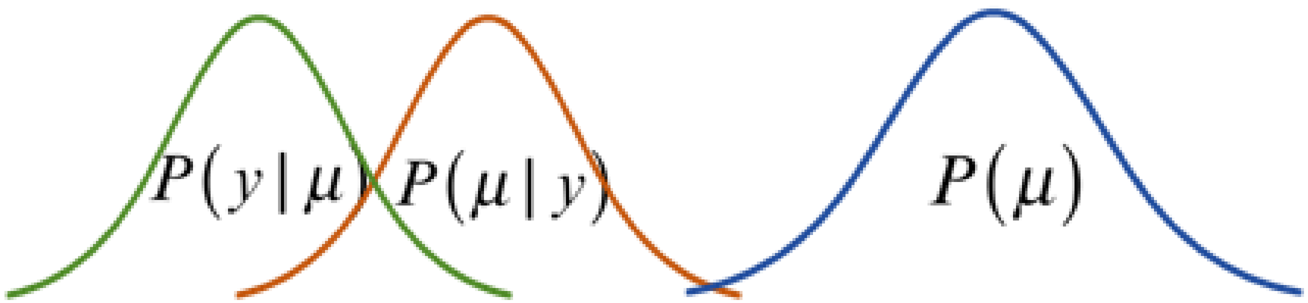
- If you want to know what the average f0 is for some speakers, you sample their f0.

- You collect data and make inferences about the mean, spread, etc.

- You cannot realistically know this ‘analytically’.

Sampling from a (Posterior) Distribution

- If you want to know about P(H|D), you sample the distribution.
- You can plot this distribution, estimate the mean, the spread, etc.
- Based on this, you can estimate probable, and improbable values of the parameter.



Example: Estimate mean of women's height

- We generate a fake sample of data: 30 observations of women's heights, in centimeters.

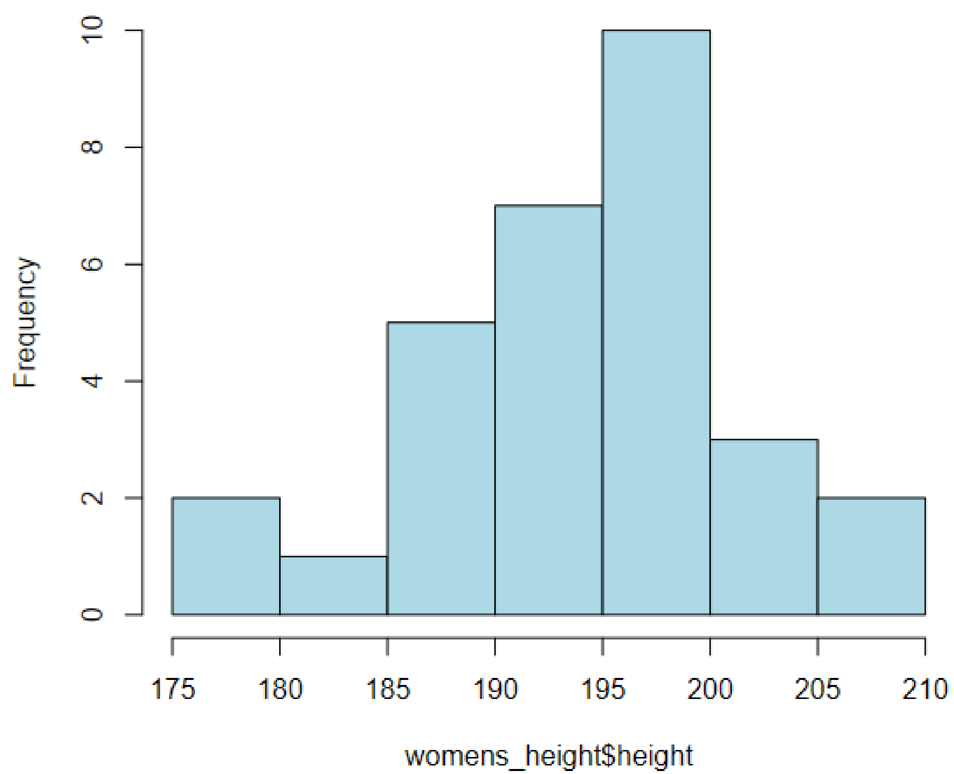
```
womens_height = data.frame (height = rnorm (30, 193, 8))
```

- The data we are creating comes from a normal distribution with a mean of 193 and a standard deviation of 8.

Example: Estimate mean of women's height

- Our data has a mean of 193.7 and a standard deviation of 7.4.

- We want to know, what is the average height of women in the USA?



Example: Bayesian model (weak prior)

- To use brms we specify a model formula, priors, and the data source.

```
library (brms)

height_model = brm (
  height ~ 1,
  data = womens_height, chains = 4, cores = 4,
  prior = c(set_prior("normal(162, 10)", class = "Intercept"))
  )
```

Example: Bayesian model (weak prior)

```
> height_model
 Family: gaussian
  Links: mu = identity; sigma = identity
Formula: height ~ 1
   Data: womens_height (Number of observations: 30)
  Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
         total post-warmup draws = 4000

Regression Coefficients:
          Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
Intercept   193.02      1.37   190.21   195.60 1.00     3235     2618

Further Distributional Parameters:
      Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sigma     7.53      1.04     5.87     9.80 1.00     2862     2327

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

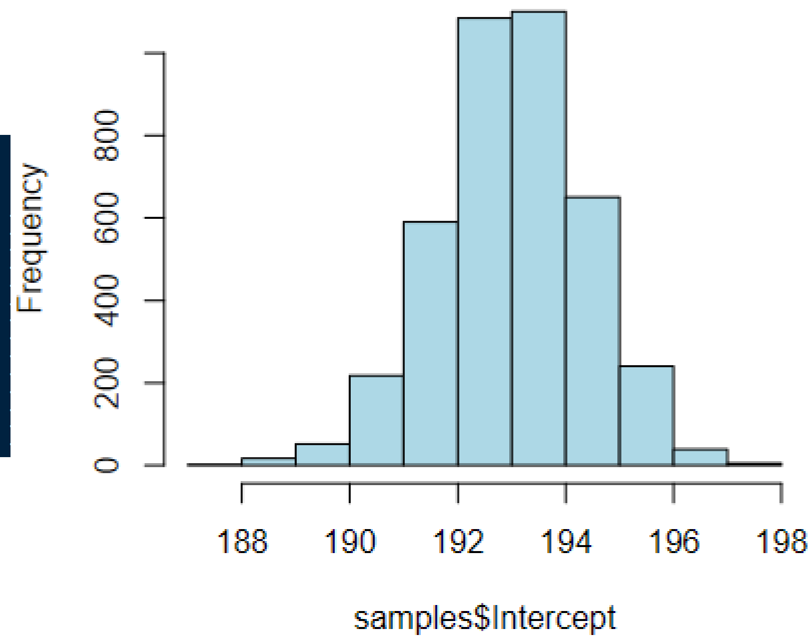
Example: Bayesian model (weak prior)

- The model output tells us about the posterior distribution of our parameters (among other things).

```
> bmmb::short_summary(height_model)
Formula:  height ~ 1
Population-Level Effects:
          Estimate Est.Error l-95% CI u-95% CI
Intercept   193.02      1.37   190.21    195.6

Family Specific Parameters:
      Estimate Est.Error l-95% CI u-95% CI
sigma     7.53      1.04     5.87      9.8
```

Posterior Samples

```
> samples = bmmb::get_samples(height_model)
> samples$Intercept
  [1] 193.1375 194.3530 191.1333 191.5903 193.9936
 [10] 192.9383 195.0827 194.5802 192.8656 194.4455
 [19] 194.1659 194.5816 195.2837 195.0103 194.2667
 [28] 193.2422 195.3528 191.8022 194.7949 193.3522
 [37] 192.9773 193.7200 192.5533 193.7762 192.9586
 [46] 193.6805 195.9992 191.5087 194.8153 191.2571
 [55] 193.1560 195.3418 194.4861 193.3837 195.4516
```




```
> mean(samples$Intercept)
[1] 193.0208
> sd(samples$Intercept)
[1] 1.365834
> quantile(samples$Intercept, c(0.025, 0.975))
    2.5%    97.5%
190.2057 195.5997
```

```
> bmmb::short_summary(height_model)
Formula:  height ~ 1
Population-Level Effects:
          Estimate Est.Error l-95% CI u-95% CI
Intercept   193.02      1.37   190.21    195.6

Family Specific Parameters:
      Estimate Est.Error l-95% CI u-95% CI
sigma     7.53      1.04     5.87      9.8
```

t-test vs. Equivalent Bayesian model

Example: one sample t-test

- Since $p < 0.05$, we reject the null hypothesis that $\mu=0$.

- We accept the alternative hypothesis that $\mu\neq0$.

```
> t.test (womens_height$height)

        One Sample t-test

data:  womens_height$height
t = 143.48, df = 29, p-value < 2.2e-16
alternative hypothesis: true mean is not equal to 0
95 percent confidence interval:
 190.8991 196.4202
sample estimates:
mean of x
 193.6597
```

Example: one sample t-test

- We use the sample mean as our value of μ.
- The 95% confidence interval will contain μ in 95% of replications.

```
> t.test (womens_height$height)

        One Sample t-test

data:  womens_height$height
t = 143.48, df = 29, p-value < 2.2e-16
alternative hypothesis: true mean is not equal to 0
95 percent confidence interval:
 190.8991 196.4202
sample estimates:
mean of x
 193.6597
```

Example: OLS Regression

```
> OLS_model = lm (height ~ 1, data = womens_height)
> summary (OLS_model)

Call:
lm(formula = height ~ 1, data = womens_height)

Residuals:
    Min      1Q  Median      3Q     Max 
-18.377  -4.139   1.393   5.010  12.103 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept)   193.66       1.35   143.5   <2e-16 ***
---
Signif. codes:  
0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 7.393 on 29 degrees of freedom
```

Comparison

t-test

```
t = 143.48, df = 29, p-value < 2.2e-16
alternative hypothesis: true mean is not equal to 0
95 percent confidence interval:
 190.8991 196.4202
sample estimates:
mean of x
 193.6597
```

OLS

```
Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   193.66       1.35   143.5   <2e-16 ***
---
```

Bayesian

```
Population-Level Effects:
          Estimate Est.Error l-95% CI u-95% CI
Intercept   193.02      1.37   190.21    195.6

Family Specific Parameters:
      Estimate Est.Error l-95% CI u-95% CI
sigma     7.53      1.04     5.87      9.8
```

Example: Bayesian model (strong prior)

- But we actually know a lot about average female height in the USA.

Table 9. Height in centimeters for females aged 20 and over and number of examined persons, selected percentiles, by race and ethnicity and age: United States, 2007–2010

Race and ethnicity and age	Number of examined persons	Mean	Standard error of the mean	5th	10th	15th	25tl
All racial and ethnic groups[1]						Centimeters	
20 years and over	5,971	162.1	0.14	150.7	153.1	154.7	157.
20–29 years	980	163.1	0.24	152.0	153.9	155.7	158.

- We can include this knowledge in our model.

```
height_model_prior = brm (
  height ~ 1,
  data = womens_height, chains = 4, cores = 4,
  prior = c(set_prior("normal(162.1, 0.14)", class = "Intercept"))
)
```

Example: Prior Comparison

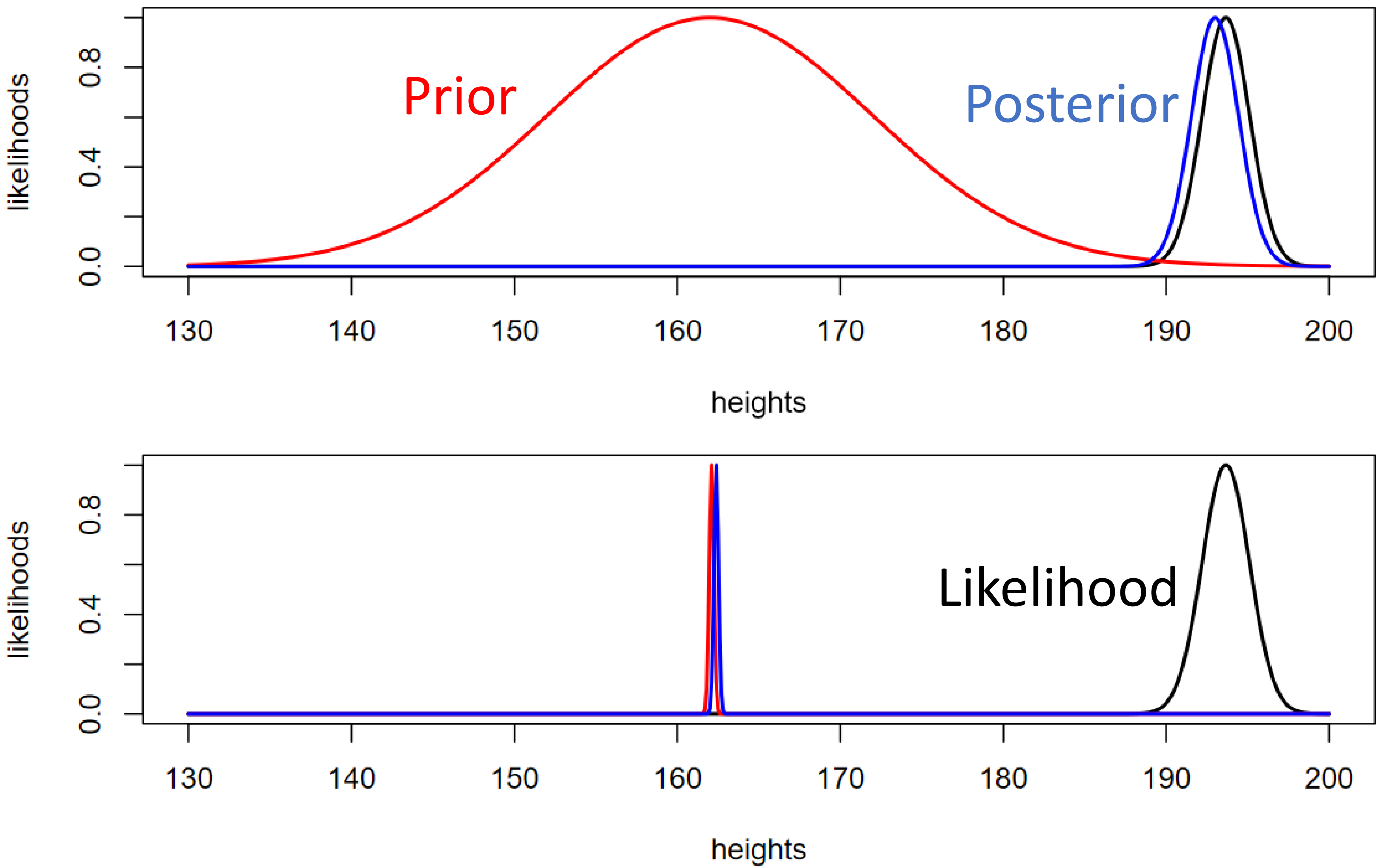
Weak Prior

```
> bmmb::short_summary(height_model)
Formula:  height ~ 1
Population-Level Effects:
          Estimate Est.Error l-95% CI u-95% CI
Intercept   193.02      1.37   190.21    195.6

Family Specific Parameters:
      Estimate Est.Error l-95% CI u-95% CI
sigma     7.53      1.04     5.87      9.8
```

Strong Prior

```
> bmmb::short_summary(height_model_prior)
Formula:  height ~ 1
Population-Level Effects:
          Estimate Est.Error l-95% CI u-95% CI
Intercept   162.12      0.14   161.85    162.4

Family Specific Parameters:
      Estimate Est.Error l-95% CI u-95% CI
sigma    31.85         4    25.19    40.95
>
```

Comparison of Prior Information



Conclusion

Bayesian Modeling

- Lets you incorporate new and old information into your work.
- Provides you with information that is more intuitive and interpretable.
- Let you make inferences based on posterior distributions, $P(H|D)$, not on likelihoods, $P(D|H)$.

brms Advantages

- Why should you use brms (and Bayesian models):

  - You can use the same syntax as lmer, and it will pick default priors for you.
  - It <u>never</u> faces convergence issues. You can <u>always</u> fit the random effect structure you want*.
  - You can fit an incredibly wide range of models using the same function.
  - You can easily fit models that are nearly impossible to fit otherwise (for the average person).

Thank You!!